

 Jal-Sanchay Tracker
Rainwater Harvesting Intelligence for Indian Households

![Version](https://img.shields.io/badge/version-1.0-blue)
![Platform](https://img.shields.io/badge/platform-Android-green)
![Kotlin](https://img.shields.io/badge/language-Kotlin-purple)
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Table of Contents

1. [Overview](#overview)
2. [Problem Statement](#problem-statement)
3. [Solution Architecture](#solution-architecture)
4. [Features](#features)
5. [Technical Stack](#technical-stack)
6. [Installation & Setup](#installation--setup)
7. [Functional Requirements](#functional-requirements)
8. [Non-Functional Requirements](#non-functional-requirements)
9. [User Flow & Experience](#user-flow--experience)
10. [Impact Goals & Success Criteria](#impact-goals--success-criteria)
11. [Feature Prioritisation](#feature-prioritisation)
12. [Future Roadmap](#future-roadmap)
13. [Project Information](#project-information)

Overview

Jal-Sanchay Tracker is a GenAI-powered Android application that transforms rainwater harvesting from an invisible habit into a **measurable, data-driven practice**. By accepting roof area, tank capacity, and daily rainfall as inputs, the app calculates **Water Wealth** in litres and converts it into relatable **Household Water Days** – giving Indian families a concrete, motivating metric for their conservation efforts.

Key Highlights

-  **100% Offline Capability** – Works without internet dependency
-  **GenAI-Powered Tips** – Personalized water-saving recommendations
-  **Intelligent Calculations** – Advanced rainwater harvesting formulas
-  **Visual Feedback** – Animated water tank dashboard
-  **Local Data Privacy** – No cloud upload, no user accounts required
-  **Designed for Non-Technical Users** – Plain language, context-aware guidance

Target Audience

- **100+ million Indian households** with existing rainwater harvesting (RWH) infrastructure
- Rural and semi-urban families seeking to optimize water conservation
- Households looking for tangible feedback on their sustainability impact

🚫 Problem Statement

The Challenge

Many households have rainwater harvesting systems installed, but **lack any way to track if they're effective**. Without data, conservation feels intangible and users quickly lose motivation to maintain their RWH systems.

Core Pain Points

- | Issue | Impact |
- |-----|-----|
- | **No Measurement Tool** | No digital tool exists to calculate actual litres collected from RWH systems |
- | **Guesswork Over Data** | Households estimate savings subjectively — no feedback loop for improvement |
- | **Lost Motivation** | Without visible impact, users abandon or neglect their RWH systems over time |
- | **Education Gap** | Concepts like 'runoff coefficient' and 'catchment area' remain unknown to average Indian households |

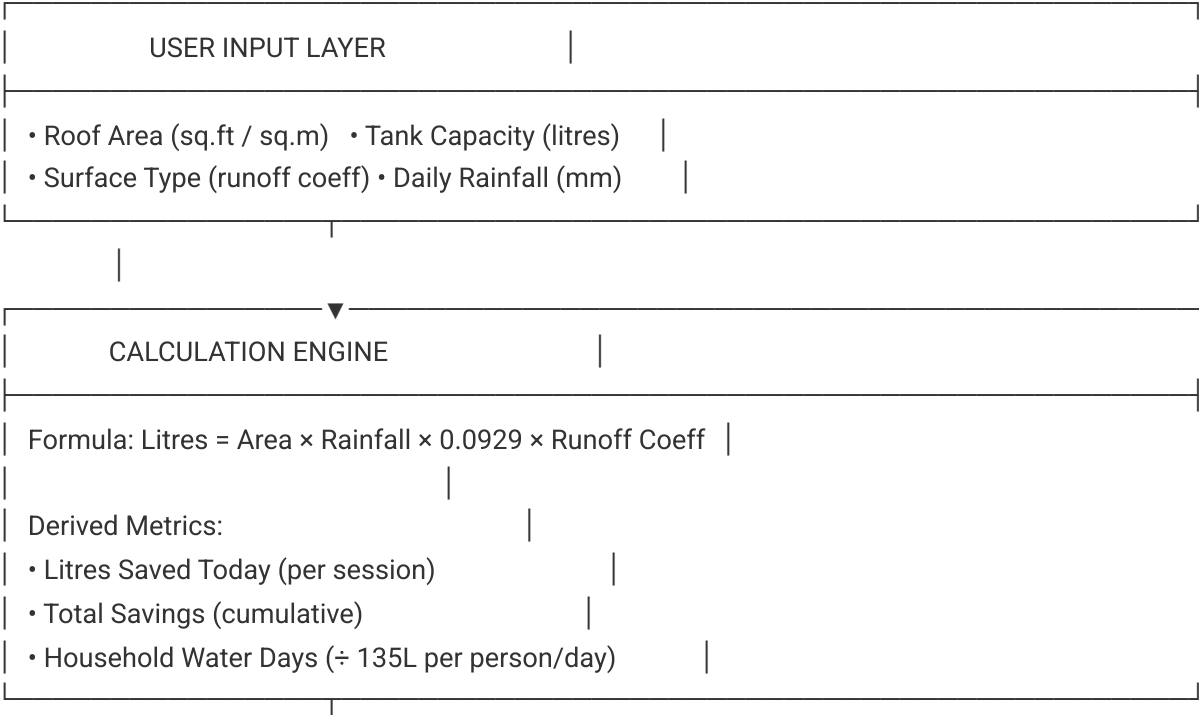
Why This Matters

- India is facing critical water scarcity challenges
- Household-level conservation requires **visible, measurable impact**
- Rural and semi-urban populations lack digital tools for resource management
- Educational awareness about water harvesting remains low

💡 Solution Architecture

How Jal-Sanchay Works

...



PRESENTATION & INTELLIGENCE LAYER	
• Animated Water Tank Dashboard	
• Monthly Reports & Summaries	
• GenAI-Powered Tips Module	
• Input Validation & Error Handling	
LOCAL DATA PERSISTENCE	
• Room Database (SQLite) – ACID-compliant local storage	
• Historical rainfall entries	
• User profile & configuration	
• Survives app crashes & device restarts	

...

✨ Features

Core Features (MVP)

- #### 1. **Setup Wizard** 🔧
- User enters roof area with unit selector (sq.ft. or sq.m.)
 - Configure tank/storage capacity in litres
 - Select surface type (Tile / Concrete / Metal Sheet)
 - Pre-defined runoff coefficients mapped to surface types
 - Data persisted locally and editable anytime

- #### 2. **Daily Rainfall Entry** 📝
- Manual input of daily rainfall in millimetres (mm)
 - Auto-tagged with current date
 - Comprehensive validation:
 - ✗ Rejects non-numeric values
 - ✗ Blocks negative or zero entries
 - ✓ Shows informative error messages
 - Edit/delete past entries functionality

3. **Water Harvesting Calculation Engine** 📊

Core Formula:

...

Litres Saved = Roof Area (sq.ft.) × Rainfall (mm) × 0.0929 × Runoff Coefficient

...

- Calculated Metrics:**
- **Litres Saved Today** — Per-session calculation
 - **Total Savings** — Cumulative total across all entries
 - **Household Water Days** — Converts abstract litres to relatable days
 - Formula: Total Savings ÷ 135L (average daily per-person use in India)

4. **Animated Water Tank Dashboard** 💧

- Visual water tank that fills proportionally to capacity
- Real-time updates on each new data entry
- Displays:
 - Litres Saved Today
 - Total Savings
 - Household Water Days
- Graceful empty state with onboarding guidance

5. **Monthly Conservation Reports** 📊

- In-app monthly summary with:
 - Total rainfall entries
 - Total litres saved
 - Average daily savings
- Shareable text summary
- Future export as PDF (v1.1+)

6. **GenAI Tips Module** 🤖

- On-demand GenAI-powered tips for better RWH practices
- Context-aware recommendations based on:
 - User's roof area
 - Usage patterns
 - Rainfall data
- Local caching; refreshed when connectivity available
- Works offline with cached tips

Additional Features (v1.0+)

Feature	Status	Priority
Input Validation & Error Handling	✅ Included	Must Have
Room DB Persistence	✅ Included	Must Have
Empty State / Onboarding	✅ Included	Must Have
Unit Toggle (sq.ft ↔ sq.m)	🔄 v1.1	Good to Have
Edit / Delete Past Entries	🔄 v1.1	Good to Have
Export / Share Report	🔄 v1.1	Good to Have
Dark Mode	🔄 v1.1	Good to Have
Runoff Coefficient Explainer	🔄 v1.0+	Good to Have

🛠️ Technical Stack

Architecture & Language

Layer	Technology
Language	Kotlin
Architecture	MVVM (Model–View–ViewModel)
Database	Room DB (SQLite wrapper)
Reactive Data	LiveData + ViewModel
UI Framework	XML Layouts + Canvas Animations
AI Module	GenAI SDK (contextual tips)

Development Tools

Tool	Purpose
IDE	Android Studio
Version Control	Git / GitHub
Min SDK	Android 8.0 (API 26)
Target SDK	Android 12+ (best practice)

```
### Key Libraries (Recommended)
kotlin

// Core Android
implementation 'androidx.core:core-ktx:1.8.0'
implementation 'androidx.appcompat:appcompat:1.4.2'

// Room Database
implementation 'androidx.room:room-runtime:2.4.2'
kapt 'androidx.room:room-compiler:2.4.2'

// Lifecycle & LiveData
implementation 'androidx.lifecycle:lifecycle-viewmodel-ktx:2.4.1'
implementation 'androidx.lifecycle:lifecycle-livedata-ktx:2.4.1'

// Material Design
implementation 'com.google.android.material:material:1.6.1'

// GenAI SDK (Anthropic/similar)
implementation 'com.anthropic:claude-sdk:1.0' // Placeholder

// Testing
testImplementation 'junit:junit:4.13.2'
androidTestImplementation 'androidx.test.espresso:espresso-core:3.4.0'
...

---
```

📦 Installation & Setup

Prerequisites

- **Android Studio** (latest version)
- **Kotlin 1.7+** support
- **Java JDK 11+**
- **Android SDK** (API 26+)

Development Environment Setup

```
```bash
```

# 1. Clone the repository

```
git clone https://github.com/mindmatrix/jal-sanchay-tracker.git
cd jal-sanchay-tracker
```

# 2. Open in Android Studio

# File → Open → Select project root

# 3. Build the project

```
./gradlew clean build
```

# 4. Run on emulator or device

```
./gradlew installDebug
```

```
```
```

App Initialization Flow

1. **First Launch**

- Splash screen with app branding
- Welcome onboarding screen
- Setup wizard activation

2. **Setup Wizard**

...

Step 1: Roof Area Input

- └─ Numeric validation
- └─ Unit selection (sq.ft / sq.m)
- └─ Save & proceed

Step 2: Tank Capacity

- └─ Litres input
- └─ Save & proceed

Step 3: Surface Type Selection

- └─ Tile (runoff: 0.6)
- └─ Concrete (runoff: 0.7)
- └─ Metal Sheet (runoff: 0.8)
- └─ Save & complete setup

...

3. ****Ready for Use****

- Dashboard displays with empty state message
- Daily entry button highlighted
- Tips section available

📋 Functional Requirements

FR-01: Setup Module

****Goal:**** Initialize user's rainwater harvesting profile

Actors: End User (household member)

Precondition: App freshly installed or setup not completed

Flow:

1. System displays Setup Wizard on app launch
2. User enters roof area with unit selector
3. User inputs tank capacity in litres
4. User selects surface type from dropdown
5. System calculates runoff coefficient from surface type
6. System validates all inputs
7. System stores profile in Room DB
8. System navigates to Dashboard

Post-condition: User profile ready; app in operational state

Error Cases:

- Non-numeric input → Show validation error
- Empty fields → Highlight required fields
- Zero or negative values → Display friendly message

FR-02: Daily Rainfall Data Entry

****Goal:**** Capture daily rainfall measurements with validation

Actors: End User

Precondition: Setup completed; user has observed rainfall

Flow:

1. User taps "Add Rainfall Entry" button
2. System shows rainfall entry dialog/form
3. User enters rainfall amount in millimetres
4. System auto-tags with current date
5. System validates input (numeric, positive, < 1000mm)
6. User confirms entry
7. System stores in Room DB
8. System triggers recalculation
9. System refreshes dashboard

Post-condition: Entry stored; dashboard updated

Error Cases:

- Non-numeric → "Please enter a valid number"
- Negative value → "Rainfall cannot be negative"
- Zero → "Enter rainfall amount greater than 0"
- Future date → "Cannot log rainfall for future dates"

...

FR-03: Water Harvesting Calculation Engine

****Goal:**** Accurately compute water savings using hydrological formula

...

Core Formula:

$$\text{Litres} = \text{Roof Area (sq.ft.)} \times \text{Rainfall (mm)} \times 0.0929 \times \text{Runoff Coefficient}$$

Derived Metrics:

- a) Litres Saved Today = Latest entry calculation
- b) Total Savings = SUM of all historical Litres
- c) Household Water Days = Total Savings ÷ 135L

Validation Rules:

- Roof Area: 1 to 10,000 sq.ft.
- Rainfall: 0.1 to 500 mm
- Runoff Coefficient: 0.6 to 0.9 (based on surface)
- Tank Capacity: 100 to 100,000 litres

Precision:

- Calculations accurate to 2 decimal places
- All results rounded to nearest whole number

...

FR-04: Dashboard Visualization

****Goal:**** Display water savings in engaging, motivational format

...

Components:

- 1. Animated Water Tank
 - Fills from 0-100% based on Total Savings vs Tank Capacity
 - Smooth animation over 800ms
 - Percentage label inside tank

2. Key Metrics Cards

| | |
|--------------------|--|
| Litres Saved Today | |
| [X] L | |

| | |
|---------------------|--|
| Total Water Savings | |
| [X,XXX] L | |

| | |
|----------------------|--|
| Household Water Days | |
| [X] days | |

3. Empty State (no data)
- Friendly message with illustration
 - "Start tracking your water wealth!"
 - Prominent "Add Rainfall Entry" CTA

Interactions:

- Pull-to-refresh → Recalculates from DB
- Tap metrics card → Show detailed breakdown
- Dashboard updates in real-time on new entries

...

FR-05: Monthly Reports

Goal: Generate consolidated conservation summaries

...

- Report Contents:
- Month & Year
 - Total rainfall entries (count)
 - Total litres saved (sum)
 - Average daily savings (mean)
 - Household water days equivalent
 - Top rainfall day (mm)

- Report Generation:
- Triggered monthly or on-demand
 - Stored in Room DB
 - Viewable in-app with detailed breakdown
 - Shareable as text summary (WhatsApp, Email, SMS)

- Future (v1.1):
- PDF export functionality
 - Charts & graphs visualization
 - Year-on-year comparison
- ...

FR-06: GenAI Tips Module

Goal: Provide intelligent, context-aware water conservation advice

...

- Features:
- On-demand tip generation (tap "Water Tips" button)
 - Context: Roof area, rainfall patterns, usage

- Response examples:
 - "Your 200 sq.ft roof catches ~20,000L/year. Try grey-water reuse!"
 - "With 15 entries, avg 10mm/day. Consider underground storage."
 - "Concrete roofs lose 10% to runoff. Clean gutters monthly!"
- Local caching (store last 10 tips)
- Refresh when internet available
- Graceful fallback to cached tips offline

Implementation:

- GenAI SDK integration (Anthropic Claude / similar)
- Max 150 tokens per response
- Max 5 API calls/day (rate limit)
- Timeout: 5 seconds (fallback to cache)

...

⚙️ Non-Functional Requirements

Performance

| Metric | Target | Justification |
|-------------------|----------------------------|---------------------------------|
| App Launch | < 2 seconds | Users expect responsive startup |
| Calculation Speed | < 100ms | Real-time dashboard updates |
| Animation FPS | 60 fps (min-range devices) | Smooth tank fill animation |
| Database Query | < 50ms | Fast data retrieval |
| UI Responsiveness | < 200ms | No lag on input or navigation |

Implementation Tips:

```

kotlin
// Use coroutines for heavy operations
viewModelScope.launch(Dispatchers.Default) {
    val result = expensiveCalculation()
    withContext(Dispatchers.Main) {
        updateUI(result)
    }
}

// Optimize Room queries with @Query annotations
@Query("SELECT SUM(litres) FROM rainfall_entries WHERE month = :month")
fun getMonthlyTotal(month: String): Flow<Long>
  
```

Offline Capability ✅

- **100% of core features** work without internet:
 - Setup & configuration
 - Data entry & validation

- Calculations & dashboard
- Monthly reports
- Historical data access
- **Only** GenAI tips module requires connectivity
- Graceful degradation: Tips show "Connect for fresh tips" when offline
- No dependency on cloud services or backend APIs

Reliability & Data Integrity

| Aspect | Requirement |
|-----------------------------|---|
| ----- ----- | |
| Data Persistence | ACID-compliant Room DB ensures no data loss |
| Crash Recovery | All historical entries recoverable after app kill |
| Device Restart | Data survives device reboot |
| Input Rollback | Failed entries don't corrupt state |
| Duplicate Prevention | Same-date entries have versioning/replacement logic |

Usability

- **Language:** Plain English (Hindi translation roadmap)
- **Terminology Explained:**
 - "Runoff Coefficient" → "% of rain that flows into tank"
 - "Catchment Area" → "Your roof area where rain lands"
- **Accessibility:**
 - Font sizes support 120%-150% scaling
 - High contrast mode compatible
 - VoiceOver/TalkBack support
- **Design:** Colorful, vibrant UI reflecting traditional Indian market aesthetics

Validation & Safety

```

````kotlin
// Comprehensive input validation
object RainfallValidator {
 fun validateRainfall(mm: String): Result<Double> {
 return when {
 mm.isEmpty() -> Result.failure(Exception("Rainfall required"))
 !mm.matches(Regex("^\\d+(\\.\\d{1,2})?$")) ->
 Result.failure(Exception("Enter valid number"))
 mm.toDouble() <= 0 ->
 Result.failure(Exception("Rainfall must be > 0"))
 mm.toDouble() > 500 ->
 Result.failure(Exception("Rainfall unrealistic"))
 else -> Result.success(mm.toDouble())
 }
 }
}

```

### Maintainability

- **Clean Architecture:** MVVM pattern enforces separation of concerns

- **Code Standards:** Kotlin naming conventions, documented APIs
- **Database Versioning:** Room migration scripts for schema updates
- **Modularity:** Features in separate packages (setup, entry, dashboard, tips)

### Security & Privacy

| Aspect                | Implementation                                 |
|-----------------------|------------------------------------------------|
| ----- -----           |                                                |
| <b>Data Storage</b>   | All data stored locally on device (no cloud)   |
| <b>User Accounts</b>  | None required – anonymous operation            |
| <b>PII Collection</b> | Zero personal information collected            |
| <b>Analytics</b>      | No tracking SDKs or telemetry                  |
| <b>Permissions</b>    | Minimal (Calendar for date, no phone/contacts) |

### Compatibility

| Requirement           | Details                                         |
|-----------------------|-------------------------------------------------|
| ----- -----           |                                                 |
| <b>Minimum API</b>    | Android 8.0 (API 26)                            |
| <b>Screen Support</b> | 5" to 7" (phones + small tablets)               |
| <b>Orientation</b>    | Portrait primary; landscape supported           |
| <b>Device Types</b>   | Smartphones primarily; tablet support secondary |
| <b>RAM</b>            | Optimized for 2GB+ devices                      |

### Scalability

Database schema designed for future expansion:

```
```sql
-- Current tables
CREATE TABLE user_profile (
  id INTEGER PRIMARY KEY,
  roof_area_sqft REAL,
  tank_capacity_liters INTEGER,
  surface_type TEXT
);

CREATE TABLE rainfall_entries (
  id INTEGER PRIMARY KEY,
  entry_date DATE,
  rainfall_mm REAL,
  litres_saved REAL
);

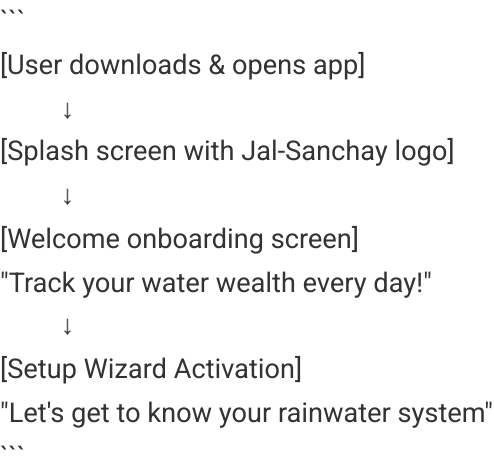
-- Future columns (backward compatible)
-- rainfall_entries: sensor_id, humidity, temperature
-- user_profile: household_id, multi_tank_setup
```

```

## 🧑🏻‍💻 User Flow & Experience

### Step-by-Step Journey

#### \*\*Step 1: Install & First Launch\*\*



#### \*\*Step 2: Setup Configuration\*\*

Setup Wizard

1

Enter Roof Area

[    sq.ft / sq.m    ]

2

Tank Capacity

[    Litres    ]

3

Surface Type

☒ Tile   ☐ Concrete   ☐ Metal

[   COMPLETE SETUP   ]

#### \*\*Step 3: Daily Workflow\*\*

Dashboard - Main Screen

 WATER TANK

45%

 Today's Savings: 85 L

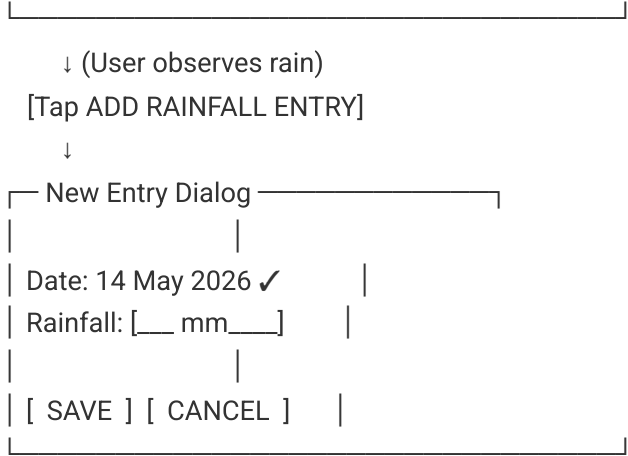
 Total Savings: 2,550 L

 Water Days: 18 days

[   + ADD RAINFALL ENTRY   ]

[    MONTHLY REPORT   ]

[    WATER TIPS   ]



↓

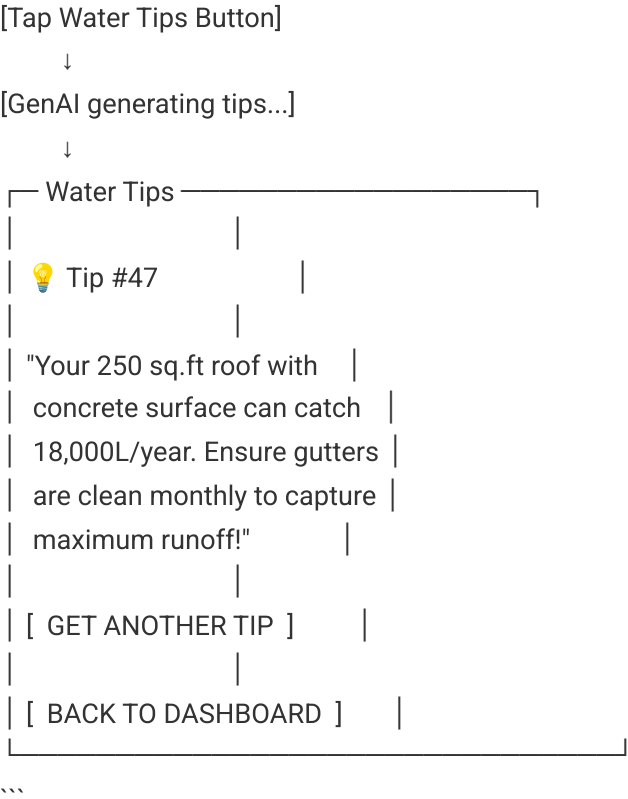
[Dashboard refreshes automatically]

"Great! You saved 125 more litres!"

...

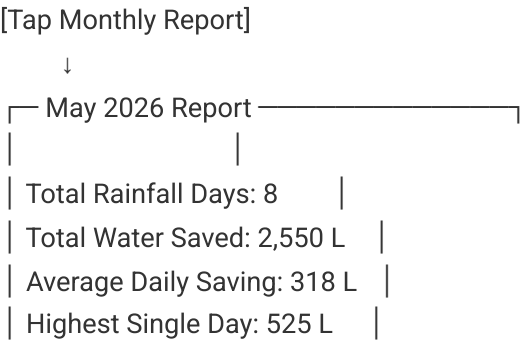
#### \*\*Step 4: Explore Tips\*\*

...



#### \*\*Step 5: View Monthly Report\*\*

...



|  |                            |  |
|--|----------------------------|--|
|  |                            |  |
|  | 🏆 Household Water Days: 18 |  |
|  |                            |  |
|  | [ SHARE SUMMARY ]          |  |
|  | [ VIEW DETAILS ]           |  |
|  |                            |  |

...

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## 🎯 Impact Goals & Success Criteria

### Impact Goals

#### 1. \*\*Water Security\*\* 💧

- Help India meet household-level water conservation targets
- Provide measurable, verifiable data for government initiatives
- Empower families to track & optimize their water usage

#### 2. \*\*Sustainability Education\*\* 📖

- Make concepts like 'Runoff Coefficient' and 'Catchment Area' real and understandable
- Build awareness of hydrological principles among ordinary citizens
- Create positive feedback loops for learning & behavior change

#### 3. \*\*Resource Management\*\* 🌱

- Enable data-driven decisions for:
  - Gardening & landscaping water allocation
  - Household usage planning
  - Water budgeting & conservation strategies
- Connect rainwater harvesting to tangible household benefits

#### 4. \*\*Behavioural Change\*\* 🔁

- Visible impact scores create positive feedback loops
- Sustained motivation for long-term conservation habits
- Community inspiration through shareable metrics

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### Success Criteria (Acceptance Tests)

|                                                                                                                                                                  |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Criterion   Test Case   Pass Condition                                                                                                                           |
| ----- ----- ---                                                                                                                                                  |
| **Tank Fill Animation**   Enter rainfall data   Water tank fills proportionally to tank capacity                                                                 |
| **Accurate Calculations**   Input roof area 100 sq.ft, rainfall 10mm, concrete surface   Calculated litres = $100 \times 10 \times 0.0929 \times 0.7 = 65.03L$ ✓ |
| **Monthly Reports**   Generate May report   Report displays total rainfall entries, litres saved, avg daily savings                                              |
| **Input Validation**   Enter non-numeric value (e.g., "abc")   System rejects gracefully with error message                                                      |
| **Offline Functionality**   Disable network; use app   All core features work (entry, calculation, dashboard, reports)                                           |

| **GenAI Tips** | Tap "Water Tips" with connectivity | At least one relevant, context-aware tip is displayed |  
| **Unit Conversion** | Enter 1000 sq.ft, toggle to sq.m | Converts correctly to ~93 sq.m; calculation adjusts accordingly |  
| **Data Persistence** | Force-kill app, restart device | All historical entries present; no data loss |  
| **Empty State** | Fresh install with no entries | Dashboard shows helpful onboarding message, not blank screen |  
| **Error Handling** | Enter -50 mm rainfall | Error message: "Rainfall cannot be negative" |

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## 🚩 Feature Prioritisation

### MoSCoW Classification

#### **Must Have** (MVP Critical)  
Features that **cannot ship without**:

| Feature                       | Rationale                               |
|-------------------------------|-----------------------------------------|
| -----                         | -----                                   |
| <b>Setup Wizard</b>           | User profile essential for calculations |
| <b>Daily Rainfall Entry</b>   | Core data input mechanism               |
| <b>Calculation Engine</b>     | Core business logic                     |
| <b>Animated Water Tank</b>    | Primary motivation/engagement hook      |
| <b>Litres Saved Today</b>     | Per-session feedback essential          |
| <b>Total Savings Counter</b>  | Running metric essential                |
| <b>Household Water Days</b>   | Relatable metric for users              |
| <b>Input Validation</b>       | Data quality & UX critical              |
| <b>Room DB Persistence</b>    | Data survival essential                 |
| <b>Empty State Onboarding</b> | UX critical for first-time users        |
| <b>Monthly Reports</b>        | Impact communication essential          |

#### **Good to Have** (v1.0+ Enhancement)  
High-value but not critical for initial release:

| Feature                      | Priority | Target Version            |
|------------------------------|----------|---------------------------|
| -----                        | -----    | ---                       |
| Unit Toggle (sq.ft ↔ sq.m)   | High     | v1.1                      |
| Edit / Delete Past Entries   | High     | v1.1                      |
| Export / Share Report        | High     | v1.1                      |
| Dark Mode                    | Medium   | v1.1                      |
| Runoff Coefficient Explainer | Medium   | v1.0+                     |
| GenAI Tips Module            | Medium   | v1.0 (if resources allow) |

#### **Won't Have** (Out of Scope v1.0)  
Features explicitly deferred:

| Feature                | Reason                                       | Target |
|------------------------|----------------------------------------------|--------|
| -----                  | -----                                        | -----  |
| Weather API Auto-Entry | Complex integration; removes user engagement | v2.0   |
| IoT Sensor Integration | Requires hardware partnerships               | v2.0   |



| Multi-Household Profiles | Adds complexity; single-family focus for v1 | v2.0 |  
| iOS / Web / Desktop | Platform expansion after Android stable | v2.0+ |  
| Village/City Dashboard | Requires backend infrastructure | v3.0 |  
| Government Data Pipeline | Policy integration; future roadmap | v3.0+ |

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## 🚀 Future Roadmap

### Version 1.1 (Q3 2026)

**\*\*Focus:\*\*** User Experience Polish & Export Capabilities

- ✅ Unit toggle (sq.ft ↔ sq.m) with automatic conversion
- ✅ Edit & delete historical entries with confirmation dialogs
- ✅ PDF report export functionality
- ✅ Share report via WhatsApp, Email, SMS
- ✅ Dark mode (following Android system theme)
- ✅ Runoff coefficient explainer (bottom sheet tooltip)
- ✅ Landscape orientation full support
- ✅ Performance optimizations & bug fixes

**\*\*Estimated Timeline:\*\*** 8-10 weeks

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### Version 2.0 (Q4 2026 - Q1 2027)

**\*\*Focus:\*\*** Automation & Intelligence

- 🔄 **\*\*Weather API Integration\*\***
  - Auto-fetch local rainfall data (OpenWeather, India Meteorological Dept)
  - Removes manual entry burden
  - Pushes notification on rain events
- 🏠 **\*\*Enhanced GenAI Module\*\***
  - More contextual tips based on seasonal patterns
  - Water-saving challenges & gamification
  - Community leaderboard (opt-in)
- 📊 **\*\*Advanced Analytics\*\***
  - Year-on-year rainfall comparison
  - Trend analysis & prediction
  - Water usage forecasting

**\*\*Estimated Timeline:\*\*** 12-14 weeks

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### Version 2.1 (Q2 2027)

**\*\*Focus:\*\* Multi-Unit & Social Features**

-  **\*\*Multi-Household Profiles\*\***
  - Support for joint families or multiple properties
  - Individual tank tracking per property
  - Family view of combined savings
-  **\*\*Community Engagement\*\***
  - Optional leaderboard (neighborhood/city)
  - Achievement badges (conservation milestones)
  - Share achievements on social media
-  **\*\*Smart Notifications\*\***
  - Daily tips at preferred time
  - Weekly summary reports
  - Conservation milestone celebrations

**\*\*Estimated Timeline:\*\* 10-12 weeks**

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**### Version 3.0 (2027+)**

**\*\*Focus:\*\* Ecosystem & Policy Integration**

-  **\*\*Village/City-Level Dashboard\*\***
  - Aggregate water conservation data (opt-in)
  - Neighborhood benchmarking
  - Municipal water security planning
-  **\*\*Government Integration\*\***
  - Water board subsidies tracking
  - Policy-level reporting pipeline
  - RWH compliance certification
-  **\*\*IoT Sensor Integration\*\***
  - Smart tank-level sensors
  - Automated real-time water level sync
  - Predictive maintenance alerts

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**##  Technical Roadmap**

...

**Version 1.0 (May 2026 - CURRENT)**

- |— Setup Module 
- |— Daily Entry 
- |— Calculation Engine 

- └─ Dashboard (animated) ✓
- └─ Monthly Reports ✓
- └─ Input Validation ✓
- └─ Room DB ✓
- └─ GenAI Tips (optional) ⚠

Version 1.1 (Q3 2026)

- └─ Unit Toggle ↺
- └─ Edit/Delete Entries ↺
- └─ Export PDF ↺
- └─ Dark Mode ↺
- └─ Coefficient Explainer ↺
- └─ Polish & Performance ↺

Version 2.0 (Q4 2026+)

- └─ Weather API 🚀
- └─ IoT Integration 🚀
- └─ Advanced Analytics 🚀
- └─ Enhanced Tips 🚀
- └─ Social Features 🚀

Version 3.0 (2027+)

- └─ Village Dashboard 🏘
- └─ Government Pipeline 🏛
- └─ IoT Ecosystem 📶
- └─ National Database 📊

...

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## 🛡 Scope Management

### In Scope (v1.0)

- ✓ Android smartphone application (5"-7" screens)
- ✓ Manual daily rainfall entry with validation
- ✓ Hydrological calculation engine
- ✓ Animated water tank visualization
- ✓ Monthly conservation reports
- ✓ GenAI-powered tips module
- ✓ 100% offline-first operation
- ✓ Room Database local persistence

### Out of Scope (Deferred to Future Versions)

- ✗ Weather API integration (v2.0)
- ✗ IoT sensor connectivity (v2.0)
- ✗ iOS, Web, or Desktop versions (v2.0+)
- ✗ Multi-household profiles (v2.0)
- ✗ Village/city dashboards (v3.0)

- ✗ Government policy reporting (v3.0+)
- ✗ Greywater or solar tracking
- ✗ User accounts & cloud sync

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## ## 📱 Screen Specifications

### ### Device Support

- **Primary:** Android phones (5"-6.5" screens)
- **Secondary:** 7" tablets
- **Orientation:** Portrait (primary), Landscape (supported)
- **Minimum API:** Android 8.0 (API 26)
- **Target API:** Android 12+ (for best practices)

### ### Screen Density Support

| Density | DPI | Device Type        |
|---------|-----|--------------------|
| LDPI    | 120 | Rare/legacy        |
| MDPI    | 160 | Base density       |
| HDPI    | 240 | Common phones      |
| XHDPI   | 320 | Most modern phones |
| XXHDPI  | 480 | High-end phones    |
| XXXHDPI | 640 | Premium devices    |

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## ## 🎨 UI/UX Design Philosophy

### ### Visual Identity

- **Color Palette:** Vibrant, warm tones reflecting traditional Indian aesthetics
  - Primary Blue: #0066CC (trust, water)
  - Accent Green: #00AA44 (growth, sustainability)
  - Warm Orange: #FF8800 (energy, culture)
- **Typography:** Clear, readable fonts optimized for non-technical users
- **Animations:** Smooth, satisfying interactions (tank fill, entry confirmation)
- **Accessibility:** High contrast, scalable fonts (120%-150%), VoiceOver support

### ### Key Design Principles

1. **Clarity over Complexity** — Every screen has one primary action
2. **Feedback Loop** — Instant visual response to user actions
3. **Motivation** — Gamify through water tank fill & milestone achievements
4. **Inclusivity** — Support varied literacy & technical backgrounds
5. **Offline-First** — Core UX doesn't depend on connectivity

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## ## 📈 Success Metrics

### ### User Engagement

- Daily Active Users (DAU)
- Monthly Active Users (MAU)
- Average session duration
- Feature usage distribution

### ### Data Quality

- Entry accuracy (validated inputs)
- Calculation validation (formula precision)
- Data retention (0% loss over device restart)

### ### Impact Metrics

- Total litres tracked across user base
- Average household water days per user
- Conservation awareness (survey feedback)

### ### Technical Metrics

- App crash rate (< 0.1%)
- Launch time (< 2 seconds)
- Database query latency (< 50ms)
- GenAI API response time (< 5 seconds)

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## ## 📋 Contributing Guidelines

### ### Development Workflow

1. **Fork & Clone:** Create feature branch from `develop`
2. **Naming Convention:** `feature/FR-XX-description` or `bugfix/issue-number`
3. **Commit Messages:** Clear, descriptive commits

...

feature: Add unit toggle for roof area input

- Implement sq.ft to sq.m conversion
- Add toggle button in setup screen
- Validate conversion accuracy

...

4. **Testing:** Unit tests for calculation engine minimum
5. **Code Review:** At least 1 approval before merge

### ### Code Standards

- Follow Kotlin naming conventions (camelCase for variables/functions)
- Use type-safe Kotlin features (avoid `!!` operator)
- Document public methods with KDoc comments
- Maintain MVVM architecture separation

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## 📞 Support & Contact

### Project Information

- \*\*Project Title:\*\* Jal-Sanchay Tracker (Project #86)
- \*\*Organization:\*\* MindMatrix VTU Internship Program
- \*\*Department:\*\* ISE (Information Science & Engineering), EPCET
- \*\*Prepared By:\*\* MindMatrix Internship Team
- \*\*Version:\*\* 1.0
- \*\*Last Updated:\*\* May 2026

### Resources

- 📄 \*\*PRD Document:\*\* [Jal\_Sanchay\_Tracker\_PRD.pdf](./Jal\_Sanchay\_Tracker\_PRD.pdf)
- 🏗️ \*\*Architecture Docs:\*\* (Link to wiki/confluence)
- 📝 \*\*Test Cases:\*\* (Link to test plan)
- 🎨 \*\*Design Mockups:\*\* (Link to Figma)

### Get Involved

- 🐛 \*\*Report Issues:\*\* GitHub Issues
- 💡 \*\*Feature Requests:\*\* GitHub Discussions
- 👥 \*\*Join Community:\*\* Slack / Discord
- ✉️ \*\*Contact:\*\* mindmatrix@example.com

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## 📄 License & Attribution

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This project is developed as part of the MindMatrix VTU Internship Program, 8th Semester, ISE Department, EPCET.

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## 🙏 Acknowledgments

This project addresses India's critical water security challenges by empowering households to track and optimize their rainwater harvesting efforts. Built with ❤️ for sustainability and community impact.

**Jal-Sanchay:** \*Conservation of Water\*

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**Last Updated:** May 14, 2026 | **Status:** Active Development | **Version:** 1.0